

HEALTHCONNECT CLINIC: ADVANCED ANALYTICS & DECISION SUPPORT REPORT

Track: Data Analytics

Project: HealthConnect Clinic Experience Lab (Week 6)

Document Type: Final Advanced Analytics Report

1. Executive Summary

Week 6 focused on transitioning exploratory data analysis into Advanced Analytics, Deep-Dive Segmentation, and Interactive Dashboard Optimization. The goal was to refine non-attendance insights, validate core operational KPIs on screen, and perform mandatory multidisciplinary integration.

Our interactive dashboard analysis confirms severe non-attendance friction driven by compounding logistical barriers and repeat behavioral patterns. The Power BI model establishes an active operational baseline of 48.5% No-Show Rate, 46.3% Attendance Rate, and 53.7% Total Wasted Capacity, representing 1,210 Lost Capacity Hours. Cross-track integration with the Data Science team confirmed that distance (>15km) and lead time (>14days) are the primary interactive drivers of appointment failure.

2. Dashboard KPI Suite

The interactive Power BI dashboard features five core operational KPI cards reflecting the live cleaned dataset:

- **No-Show Rate % (48.5%):** Displays the primary operational drop-off baseline across total scheduled clinic appointments.
- **Attendance Rate % (46.3%):** Measures actual realized clinic capacity and patient throughput.
- **Wasted Capacity % (53.7%):** Quantifies total unutilized appointment slots combining no-shows (48.5%) and late cancellations (5.2%).
- **Lost Capacity Hours (1.21K / 1,210 Hours):** Translates non-attendance into direct unutilized provider time based on 30-minute appointment slots.
- **SMS No-Show Rate % (45.8%):** Identifies SMS as the top-performing reminder channel, achieving a ~5.6% lower non-attendance rate compared to unreminded slots (51.4%).

3. Deep-Dive Segmentation & Advanced Findings

- **Compounding Logistical Friction (>15km * >14Days):** Appointments with lead times exceeding 14 days and travel distances over 15 km experience non-attendance rates between **44.2% and 57.0%**, more than double local short-lead baselines.
- **Age Cohort Transport Barriers:** Elderly patients (≥ 65 years) living >10 km away suffer a 38.5% no-show rate, driven by transportation logistics rather than appointment forgetfulness.
- **Channel Efficiency:** SMS notifications reduce non-attendance by 34% among patients aged 18–45, but show less than 5% impact for patients aged 65+, indicating a need for channel-stratified outreach.

4. Dashboard Architecture & Visual Design

The dashboard utilizes a 16:9 structured layout with a sidebar navigation panel and clear diagnostic flow:

- **Top Panel:** Executive KPI Cards with visual alert formatting (Red for Wasted Capacity, Green for SMS Performance).
- **Middle Section:**
 - *Lead Time Curve (Line Chart):* Tracks non-attendance escalation as booking lead days increase from 0 to 60.
 - *Channel Performance (Clustered Bar Chart):* Compares no-show rates across SMS (45.8%), Email, WhatsApp, and None (51.4%).
 - *Prior History Impact (Donut Chart):* Displays appointment distribution by prior no-shows (0, 1, 2, 3+), proving chronic non-attenders account for 36.5% of misses.
- **Bottom Section:**
 - *Compounding Risk Matrix:* Cross-tabulates Lead Time Buckets against Distance Tiers (<5km, 5–15km, >15km) with soft red conditional formatting.
 - *Actionable Operational Queue (Table Visual):* Filters upcoming appointments with `Lead Time > 14 Days` and `Reminder Sent = False` for targeted administrative outreach.

5. Key Operational Recommendations

- **Telehealth Triage for Distant Patients:** Offer virtual consultations for non-urgent appointments where patient distance exceeds 15 km.
- **Age-Stratified Reminder Protocols:** Maintain automated SMS cadences for patients under 45 while shifting patients ≥ 65 to direct phone calls and transport support.
- **Dynamic Waitlist Recovery:** Deploy automated waitlist backfilling for unconfirmed appointments with lead times >14 days to recover lost provider hours.

6. Cross-Track Integration (Data Science Collaboration)

Integration Element	Details & Evidence
Track Collaborated With	Data Science Track
Data Provided to Data Science	Segment interaction findings (Distance >15km X Lead Time >14 days), binned risk flags, and KPI correlation matrices.
Data Received from Data Science	Feature importance rankings (Random Forest/Gradient Boosting) and baseline error distributions (Lead time bias in false positives).
Integration Outcome	Data Science validated our segment findings by incorporating interaction terms into their refined Gradient Boosting candidate model, reducing false positives from 191 to 175.

7. Week 7 Testing Requirements

- Validate interactive dashboard slicers against administrative user workflows.
- Test dynamic waitlist backfill logic to measure theoretical capacity recovery.
- Conduct fairness and access audits across age cohorts and geographic zones.